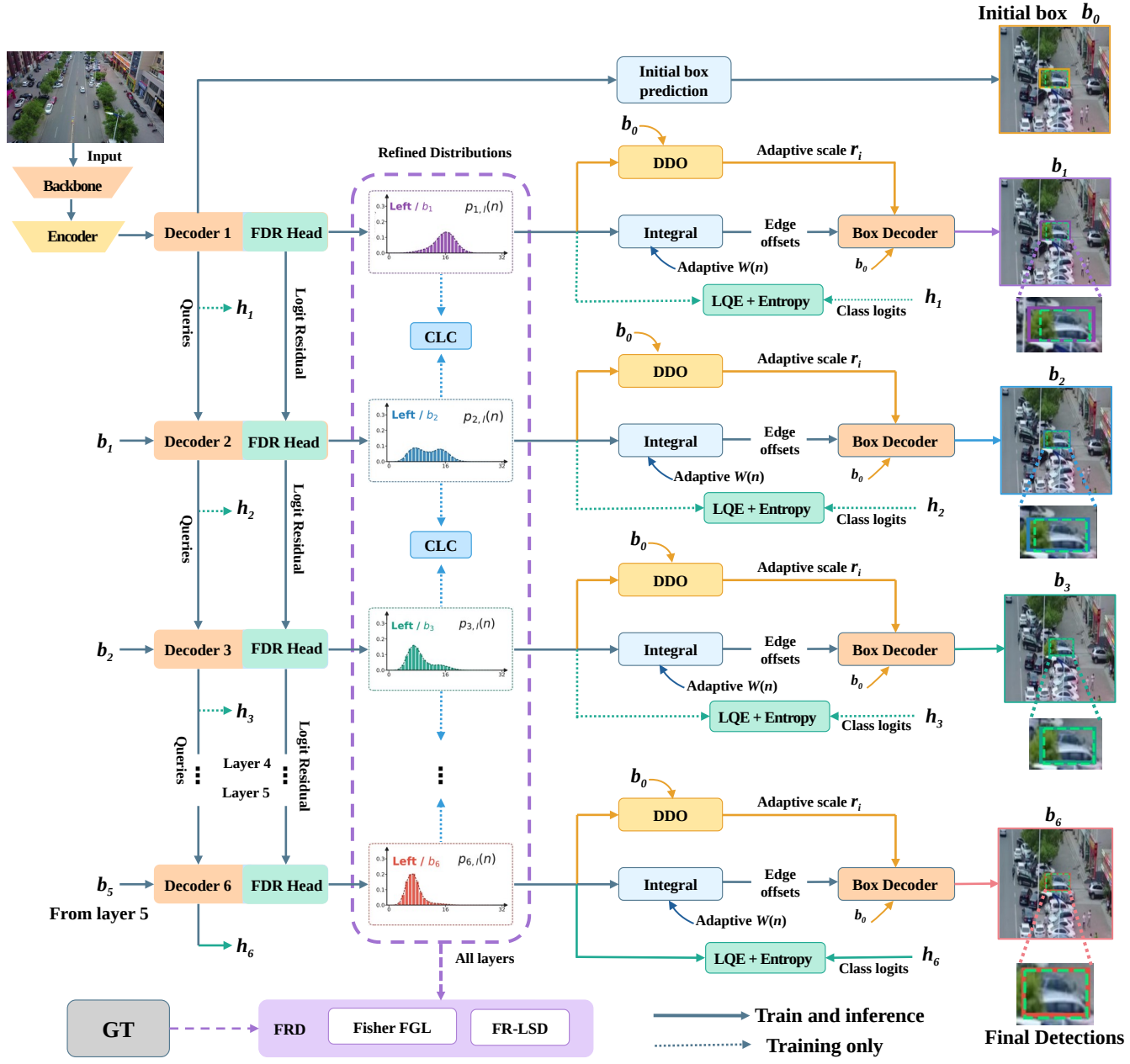
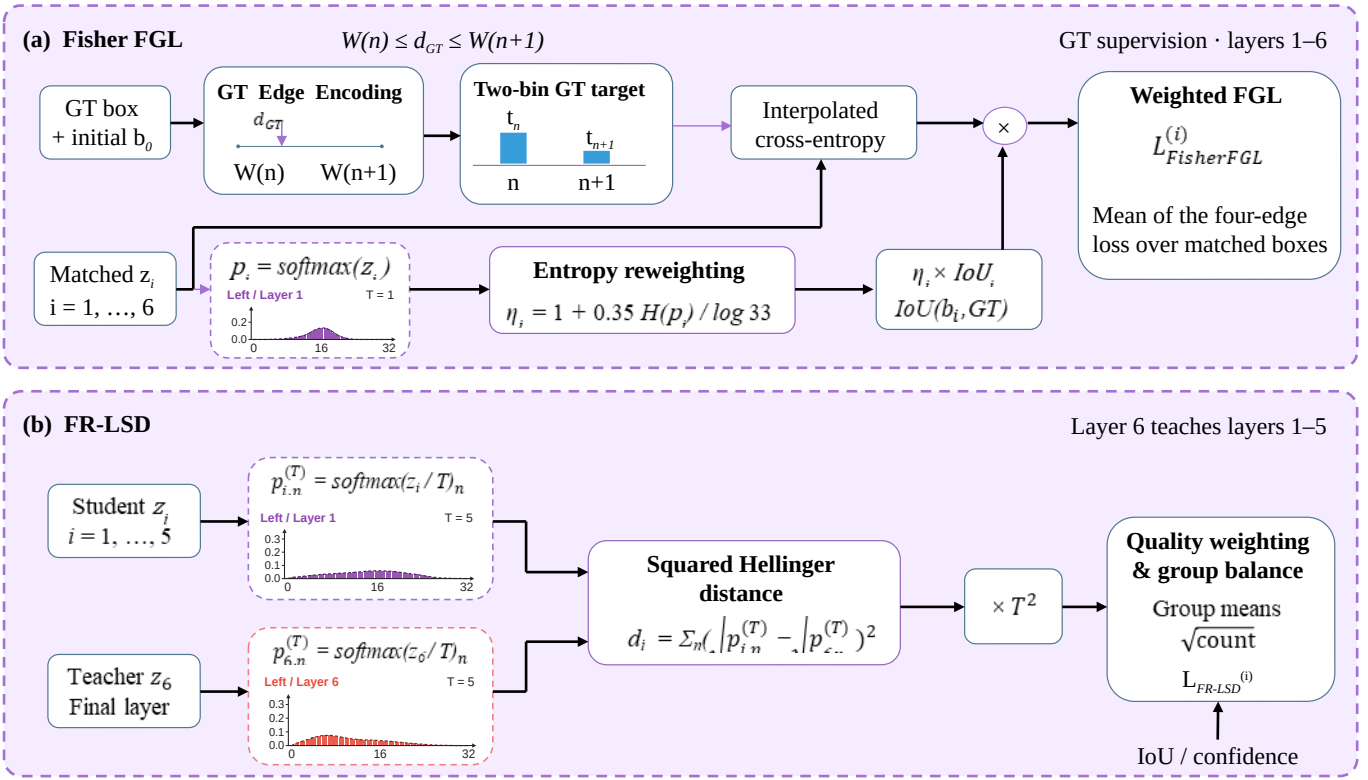
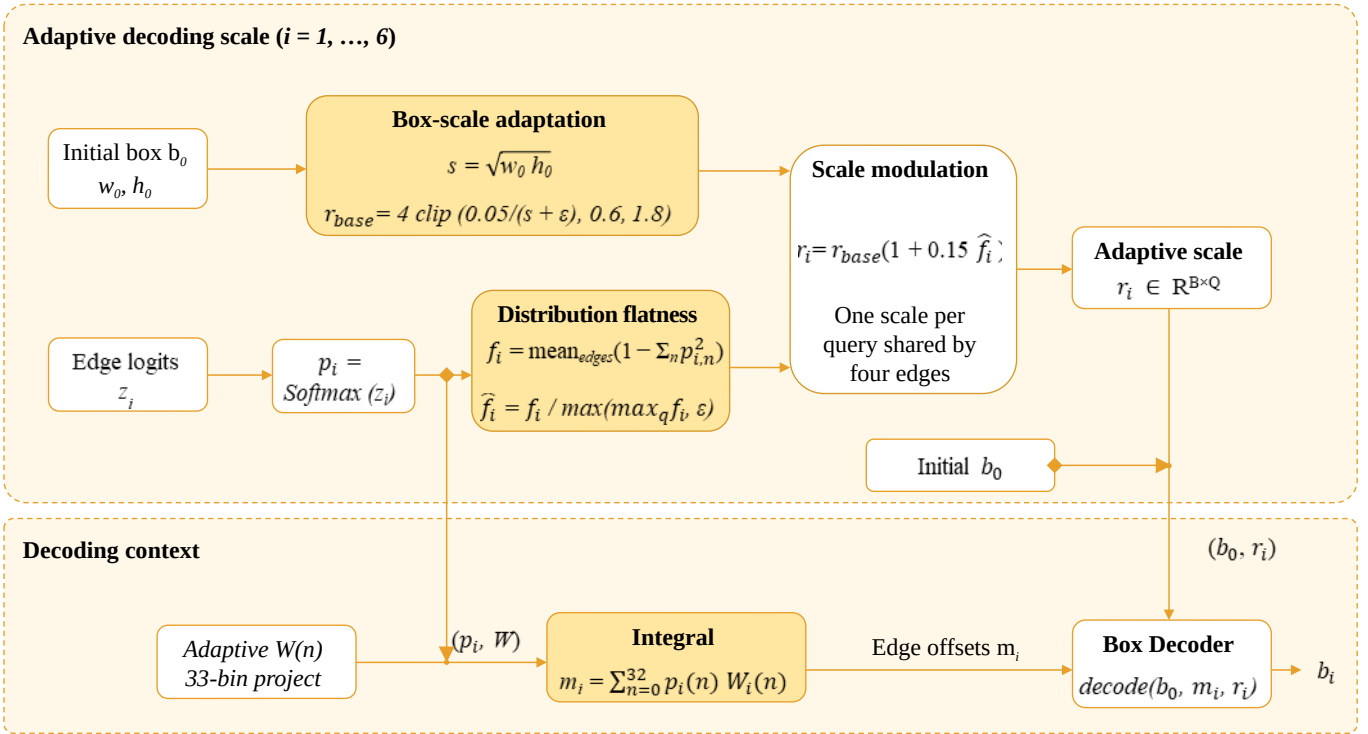

Initial box b_0
Input
Backbone
Encoder
Initial box prediction
Refined Distributions
Decoder 1
FDR Head
Queries
h_1
Logit Residual
CLC
b_1
Decoder 2
h_2
b_2
Decoder 3
h_3
Layer 4
Layer 5
b_5
From layer 5
Decoder 6
h_6
All layers
Left / b_1 $p_{1,i}(n)$
Left / b_2 $p_{2,i}(n)$
Left / b_3 $p_{3,i}(n)$
Left / b_6 $p_{6,i}(n)$
DDO
Adaptive scale r_i
Integral
Edge offsets
Box Decoder
Adaptive $W(n)$
LQE + Entropy
Class logits
b_0
b_1
b_2
b_3
b_6
h_1
h_2
h_3
h_6
GT
FRD
Fisher FGL
FR-LSD
Train and inference
Training only
Final Detections

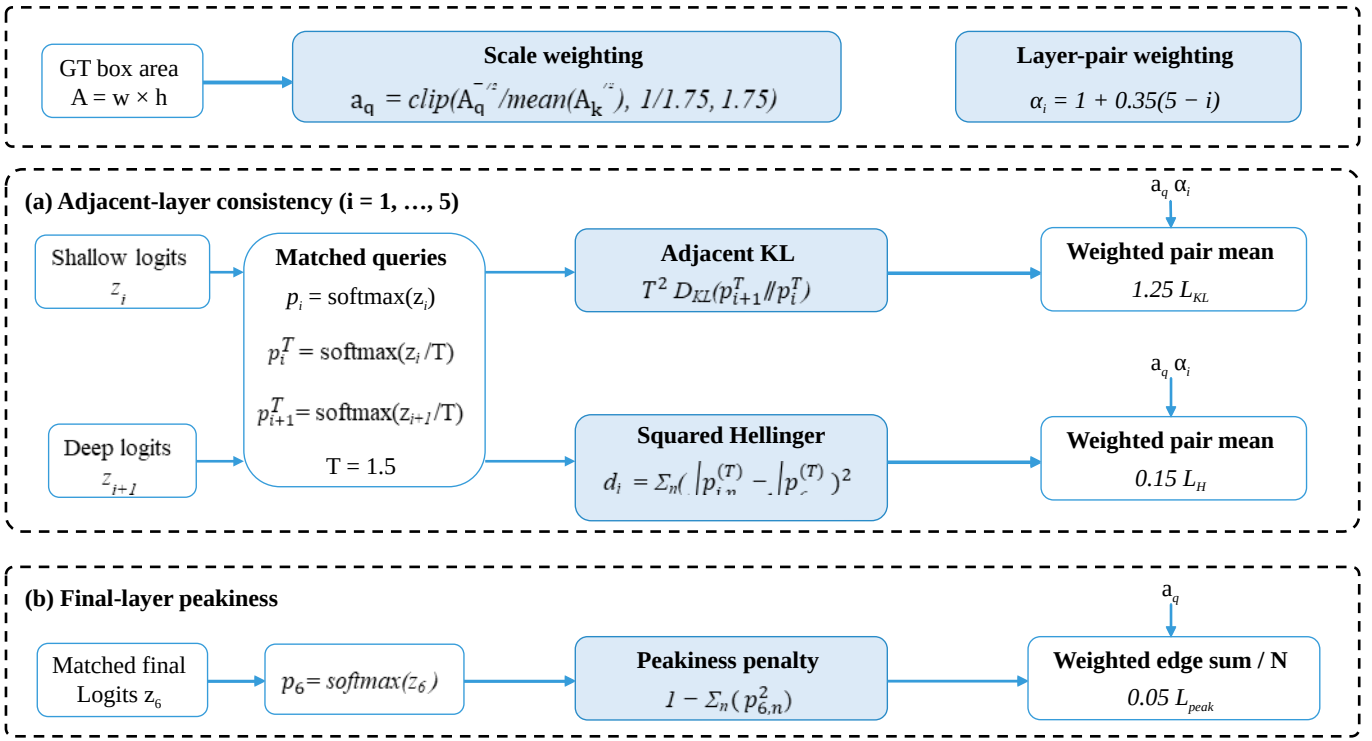
Fisher–Rao Distribution Loss



Decoder Distribution Optimizer (DDO)



Cross-Layer Consistency



Localization Quality Estimator (LQE)

